

# HIGH-DIMENSIONAL CONTINUOUS CONTROL USING GENERALIZED ADVANTAGE ESTIMATION

—— ICLR 2016

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- The estimation of return/value
- The estimation of advantage
- Experiments
- From a formal point of view
- The relationship with reward shaping

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## Value Based Methods

Policy Evaluation (Prediction)

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s]$$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{T-1} R_T$$

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a]$$

Monte Carlo Prediction

$$V(S_t) \leftarrow V(S_t) + \alpha[G_t - V(S_t)]$$

TD Prediction

$$V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

$$n = 1 \text{ (TD(0))} \quad G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1})$$

$$n = 2 \quad G_t^{(2)} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$$

.....

$$n = \infty \text{ (MC)} \quad G_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-t-1} R_T$$



$$\text{n-step return: } G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

$$\text{n-step TD: } V(S_t) \leftarrow V(S_t) + \alpha (G_t^{(n)} - V(S_t))$$

1-step TD  
and TD(0)



2-step TD



3-step TD



...

n-step TD



$\infty$ -step TD  
and Monte Carlo



High bias and low variance



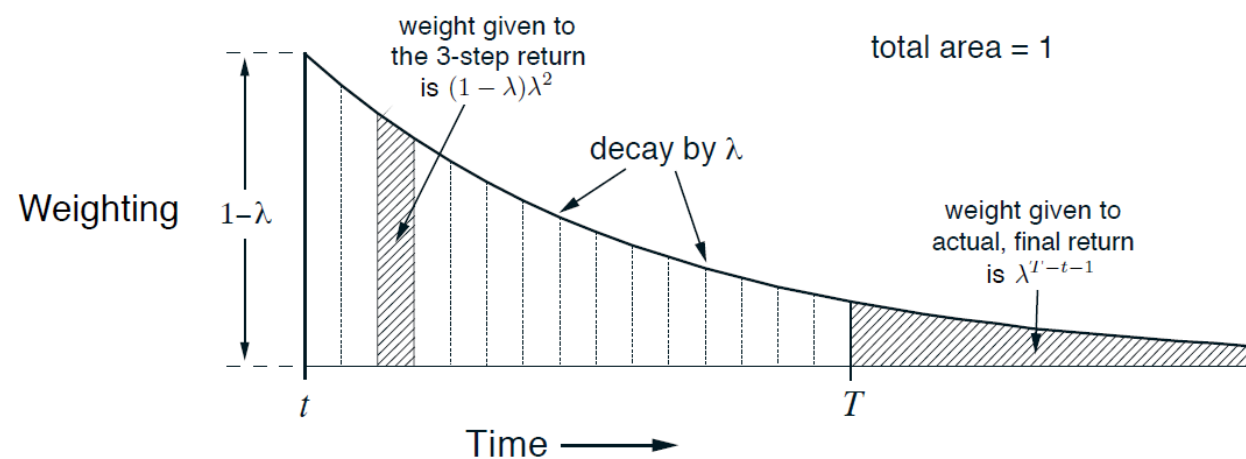
Low bias and high variance

Trade-off between bias and variance

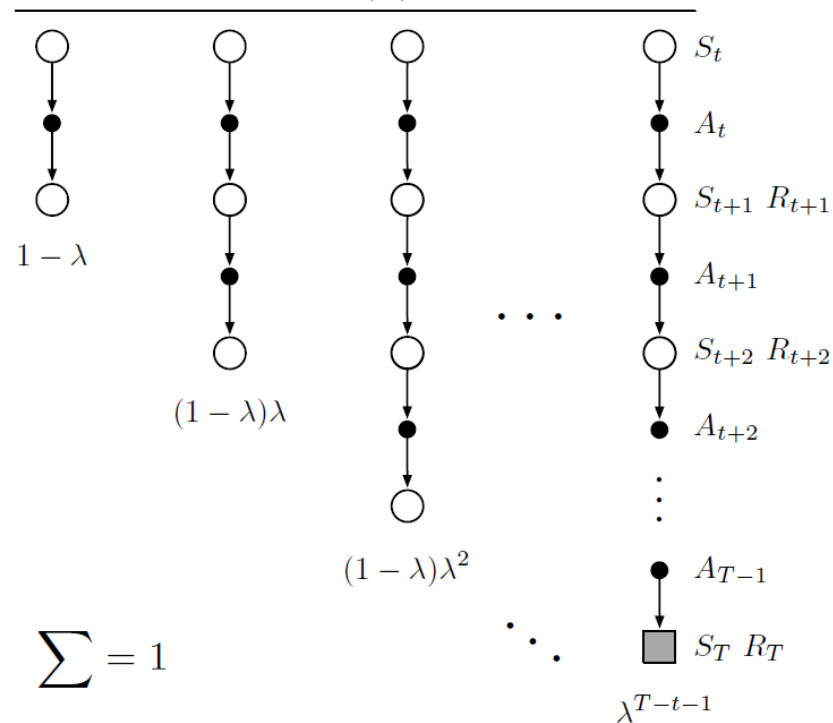
$\lambda$ -return

$$G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$$

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$



TD( $\lambda$ )



$$\lambda = 0, G_t^\lambda = G_t^{(1)} = R_{t+1} + \gamma V(S_{t+1}) \quad TD(0)$$

$$\lambda = 1, G_t^\lambda = G_t^{(\infty)} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-t-1} R_T \quad MC$$

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## Policy Gradient Methods

REINFORCE: Monte Carlo Policy Gradient

$$\boldsymbol{\theta}_{t+1} \doteq \boldsymbol{\theta}_t + \alpha G_t \frac{\nabla \pi(A_t | S_t, \boldsymbol{\theta}_t)}{\pi(A_t | S_t, \boldsymbol{\theta}_t)}.$$

REINFORCE with Baseline

$$\boldsymbol{\theta}_{t+1} \doteq \boldsymbol{\theta}_t + \alpha \left( G_t - b(S_t) \right) \frac{\nabla \pi(A_t | S_t, \boldsymbol{\theta}_t)}{\pi(A_t | S_t, \boldsymbol{\theta}_t)}.$$



Advantage function

Actor–Critic Methods

$$\begin{aligned} \boldsymbol{\theta}_{t+1} &\doteq \boldsymbol{\theta}_t + \alpha \left( G_{t:t+1} - \hat{v}(S_t, \mathbf{w}) \right) \frac{\nabla \pi(A_t | S_t, \boldsymbol{\theta}_t)}{\pi(A_t | S_t, \boldsymbol{\theta}_t)} \\ &= \boldsymbol{\theta}_t + \alpha \left( R_{t+1} + \gamma \hat{v}(S_{t+1}, \mathbf{w}) - \hat{v}(S_t, \mathbf{w}) \right) \frac{\nabla \pi(A_t | S_t, \boldsymbol{\theta}_t)}{\pi(A_t | S_t, \boldsymbol{\theta}_t)} \\ &= \boldsymbol{\theta}_t + \alpha \delta_t \frac{\nabla \pi(A_t | S_t, \boldsymbol{\theta}_t)}{\pi(A_t | S_t, \boldsymbol{\theta}_t)}. \end{aligned}$$



TD in Policy Gradient

Advantage function

$$A^{\pi, \gamma}(s_t, a_t) = Q^{\pi, \gamma}(s_t, a_t) - V^{\pi, \gamma}(s_t)$$

$$A^{\pi, \gamma}(s_t, a_t) = r_t + \gamma V^{\pi, \gamma}(s_{t+1}) - V^{\pi, \gamma}(s_t)$$

$$\begin{aligned}\hat{A}_t^{(1)} &:= \delta_t^V &= -V(s_t) + r_t + \gamma V(s_{t+1}) \\ \hat{A}_t^{(2)} &:= \delta_t^V + \gamma \delta_{t+1}^V &= -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 V(s_{t+2}) \\ \hat{A}_t^{(3)} &:= \delta_t^V + \gamma \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V &= -V(s_t) + r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \gamma^3 V(s_{t+3})\end{aligned}$$

$$\hat{A}_t^{(k)} := \sum_{l=0}^{k-1} \gamma^l \delta_{t+l}^V = -V(s_t) + r_t + \gamma r_{t+1} + \cdots + \gamma^{k-1} r_{t+k-1} + \gamma^k V(s_{t+k})$$

$$\hat{A}_t^{(\infty)} = \sum_{l=0}^{\infty} \gamma^l \delta_{t+l}^V = -V(s_t) + \sum_{l=0}^{\infty} \gamma^l r_{t+l}$$

**Combine all of these.**

$$\begin{aligned}
\hat{A}_t^{\text{GAE}(\gamma, \lambda)} &:= (1 - \lambda) \left( \hat{A}_t^{(1)} + \lambda \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \dots \right) \\
&= (1 - \lambda) \left( \delta_t^V + \lambda(\delta_t^V + \gamma \delta_{t+1}^V) + \lambda^2(\delta_t^V + \gamma \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V) + \dots \right) \\
&= (1 - \lambda) \left( \delta_t^V (1 + \lambda + \lambda^2 + \dots) + \gamma \delta_{t+1}^V (\lambda + \lambda^2 + \lambda^3 + \dots) \right. \\
&\quad \left. + \gamma^2 \delta_{t+2}^V (\lambda^2 + \lambda^3 + \lambda^4 + \dots) + \dots \right) \\
&= (1 - \lambda) \left( \delta_t^V \left( \frac{1}{1 - \lambda} \right) + \gamma \delta_{t+1}^V \left( \frac{\lambda}{1 - \lambda} \right) + \gamma^2 \delta_{t+2}^V \left( \frac{\lambda^2}{1 - \lambda} \right) + \dots \right) \\
&= \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V
\end{aligned} \tag{16}$$

There are two notable special cases of this formula, obtained by setting  $\lambda = 0$  and  $\lambda = 1$ .

$$\text{GAE}(\gamma, 0) : \quad \hat{A}_t := \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{17} \quad \text{High bias (TD(0))}$$

$$\text{GAE}(\gamma, 1) : \quad \hat{A}_t := \sum_{l=0}^{\infty} \gamma^l \delta_{t+l} = \sum_{l=0}^{\infty} \gamma^l r_{t+l} - V(s_t) \tag{18} \quad \text{High variance (MC)}$$

Using the generalized advantage estimator, we can construct a biased estimator of  $g^\gamma$ , the discounted policy gradient from Equation (6):

$$g^\gamma \approx \mathbb{E} \left[ \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \hat{A}_t^{\text{GAE}(\gamma, \lambda)} \right] = \mathbb{E} \left[ \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V \right], \tag{19}$$

where equality holds when  $\lambda = 1$ .

How to use it?

MC + GAE

$$\hat{V}_t = \sum_{l=0}^{\infty} \gamma^l r_{t+l}$$

$$\hat{A}_t^{GAE(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V \quad \gamma = 0.99, \lambda = 0.95$$

Why not  $TD(\lambda)$ ?

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<sup>2</sup>Another natural choice is to compute target values with an estimator based on the  $TD(\lambda)$  backup (Bertsekas, 2012; Sutton & Barto, 1998), mirroring the expression we use for policy gradient estimation:  $\hat{V}_t^\lambda = V_{\phi_{\text{old}}}(s_t) + \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}$ . While we experimented with this choice, we did not notice a difference in performance from the  $\lambda = 1$  estimator in Equation (28).

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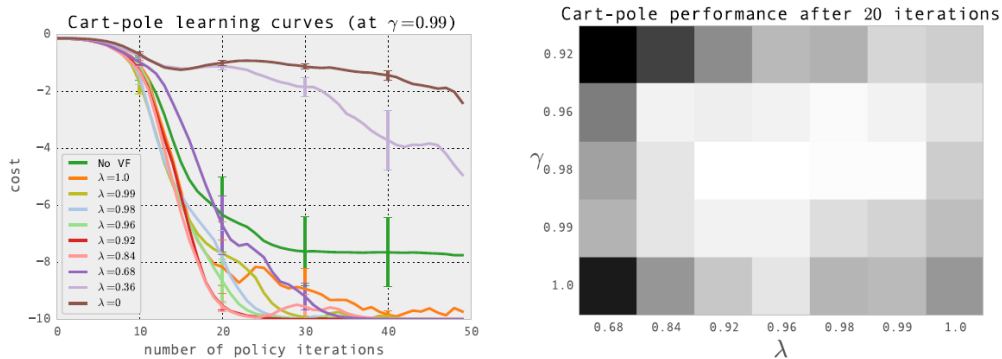


Figure 2: Left: learning curves for cart-pole task, using generalized advantage estimation with varying values of  $\lambda$  at  $\gamma = 0.99$ . The fastest policy improvement is obtain by intermediate values of  $\lambda$  in the range  $[0.92, 0.98]$ . Right: performance after 20 iterations of policy optimization, as  $\gamma$  and  $\lambda$  are varied. White means higher reward. The best results are obtained at intermediate values of both.

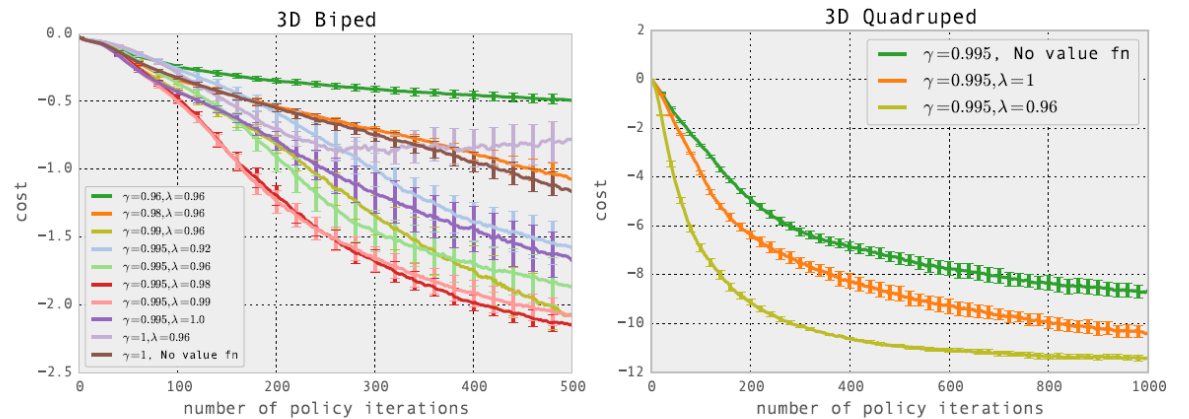


Figure 3: Left: Learning curves for 3D bipedal locomotion, averaged across nine runs of the algorithm. Right: learning curves for 3D quadrupedal locomotion, averaged across five runs.

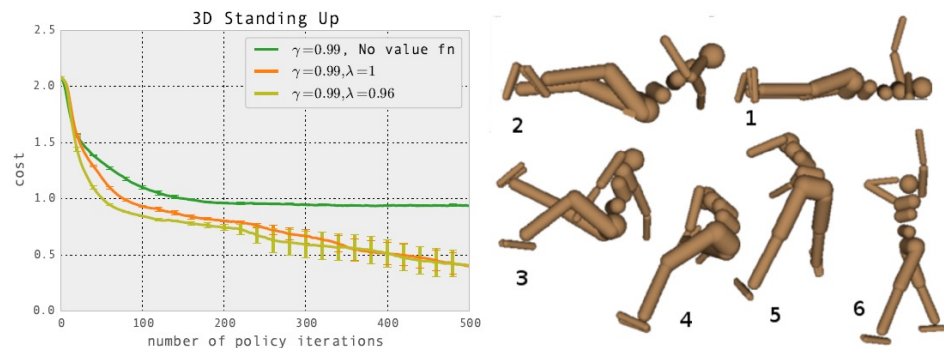


Figure 4: (a) Learning curve from quadrupedal walking, (b) learning curve for 3D standing up, (c) clips from 3D standing up.

## Atari game: Alien

| Configuration | Last-iteration Mean Score            | Difference with BASELINE |
|---------------|--------------------------------------|--------------------------|
| BASELINE      | <b>1411.917</b> [1296.935, 1526.898] | -                        |
| PPO-M         | <b>421.79</b> [393.561, 450.019]     | <b>-70.13%</b>           |
| PPO-M-W       | <b>1042.968</b> [966.957, 1118.980]  | <b>-26.13%</b>           |
| PPO-NoV       | <b>1103.503</b> [1037.946, 1169.061] | <b>-21.84%</b>           |
| PPO-NoR       | <b>774.5233</b> [721.3818, 827.6649] | <b>-45.14%</b>           |
| PPO-X         | <b>1137.52</b> [1004.209, 1270.831]  | <b>-19.43%</b>           |
| PPO-P         | <b>1266.7</b> [1191.503, 1341.897]   | <b>-10.29%</b>           |

Table 2: Comparison among different configurations at the default learning rate. All results are averaged over 30 runs with different random seeds. The last-iteration mean score with 95% confidence interval is shown in table.

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Policy gradient methods maximize the expected total reward by repeatedly estimating the gradient  $g := \nabla_{\theta} \mathbb{E} [\sum_{t=0}^{\infty} r_t]$ . There are several different related expressions for the policy gradient, which have the form

$$g = \mathbb{E} \left[ \sum_{t=0}^{\infty} \Psi_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right], \quad (1)$$

where  $\Psi_t$  may be one of the following:

1.  $\sum_{t=0}^{\infty} r_t$ : total reward of the trajectory.
2.  $\sum_{t'=t}^{\infty} r_{t'}$ : reward following action  $a_t$ .
3.  $\sum_{t'=t}^{\infty} r_{t'} - b(s_t)$ : baselined version of previous formula.
4.  $Q^{\pi}(s_t, a_t)$ : state-action value function.
5.  $A^{\pi}(s_t, a_t)$ : advantage function.
6.  $r_t + V^{\pi}(s_{t+1}) - V^{\pi}(s_t)$ : TD residual.

The latter formulas use the definitions

$$V^{\pi}(s_t) := \mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} \left[ \sum_{l=0}^{\infty} r_{t+l} \right] \quad Q^{\pi}(s_t, a_t) := \mathbb{E}_{s_{t+1:\infty}, a_{t+1:\infty}} \left[ \sum_{l=0}^{\infty} r_{t+l} \right] \quad (2)$$

$$A^{\pi}(s_t, a_t) := Q^{\pi}(s_t, a_t) - V^{\pi}(s_t), \quad (\text{Advantage function}). \quad (3)$$

The choice  $\Psi_t = A^{\pi}(s_t, a_t)$  yields almost the lowest possible variance.

The discounted approximation to the policy gradient is defined as follows:

$$g^\gamma := \mathbb{E}_{\substack{s_{0:\infty} \\ a_{0:\infty}}} \left[ \sum_{t=0}^{\infty} A^{\pi,\gamma}(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right]. \quad (6)$$

TODO: obtain biased (but not too biased) estimators for  $A^{\pi,\gamma}$ .

**Definition 1.** *The estimator  $\hat{A}_t$  is  $\gamma$ -just if*

$$\mathbb{E}_{\substack{s_{0:\infty} \\ a_{0:\infty}}} \left[ \hat{A}_t(s_{0:\infty}, a_{0:\infty}) \nabla_\theta \log \pi_\theta(a_t | s_t) \right] = \mathbb{E}_{\substack{s_{0:\infty} \\ a_{0:\infty}}} [A^{\pi,\gamma}(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t)]. \quad (7)$$

It follows immediately that if  $\hat{A}_t$  is  $\gamma$ -just for all  $t$ , then

$$\mathbb{E}_{\substack{s_{0:\infty} \\ a_{0:\infty}}} \left[ \sum_{t=0}^{\infty} \hat{A}_t(s_{0:\infty}, a_{0:\infty}) \nabla_\theta \log \pi_\theta(a_t | s_t) \right] = g^\gamma \quad (8)$$

The expressions are all  $\gamma$ -just advantage estimators for  $\hat{A}_t$ :

- $\sum_{l=0}^{\infty} \gamma^l r_{t+l}$
- $Q^{\pi,\gamma}(s_t, a_t)$
- $A^{\pi,\gamma}(s_t, a_t)$
- $r_t + \gamma V^{\pi,\gamma}(s_{t+1}) - V^{\pi,\gamma}(s_t)$ .

**Proposition 1.** Suppose that  $\hat{A}_t$  can be written in the form  $\hat{A}_t(s_{0:\infty}, a_{0:\infty}) = Q_t(s_{t:\infty}, a_{t:\infty}) - b_t(s_{0:t}, a_{0:t-1})$  such that for all  $(s_t, a_t)$ ,  $\mathbb{E}_{s_{t+1:\infty}, a_{t+1:\infty} \mid s_t, a_t} [Q_t(s_{t:\infty}, a_{t:\infty})] = Q^{\pi, \gamma}(s_t, a_t)$ . Then  $\hat{A}$  is  $\gamma$ -just.

**Proof of Proposition 1:** First we can split the expectation into terms involving  $Q$  and  $b$ ,

$$\begin{aligned} & \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) (Q_t(s_{0:\infty}, a_{0:\infty}) - b_t(s_{0:t}, a_{0:t-1}))] \\ &= \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) (Q_t(s_{0:\infty}, a_{0:\infty}))] \\ & \quad - \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) (b_t(s_{0:t}, a_{0:t-1}))] \end{aligned} \tag{33}$$

We'll consider the terms with  $Q$  and  $b$  in turn.

$$\begin{aligned} & \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) Q_t(s_{0:\infty}, a_{0:\infty})] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t}} [\mathbb{E}_{s_{t+1:\infty}, a_{t+1:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) Q_t(s_{0:\infty}, a_{0:\infty})]] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) \mathbb{E}_{s_{t+1:\infty}, a_{t+1:\infty}} [Q_t(s_{0:\infty}, a_{0:\infty})]] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) A^{\pi}(s_t, a_t)] \end{aligned}$$

Next,

$$\begin{aligned} & \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) b_t(s_{0:t}, a_{0:t-1})] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [\mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) b_t(s_{0:t}, a_{0:t-1})]] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [\mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} [\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t)] b_t(s_{0:t}, a_{0:t-1})] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [0 \cdot b_t(s_{0:t}, a_{0:t-1})] \\ &= 0. \end{aligned}$$

$$\begin{aligned}
\hat{A}_t^{\text{GAE}(\gamma, \lambda)} &:= (1 - \lambda) \left( \hat{A}_t^{(1)} + \lambda \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \dots \right) \\
&= (1 - \lambda) \left( \delta_t^V + \lambda(\delta_t^V + \gamma \delta_{t+1}^V) + \lambda^2(\delta_t^V + \gamma \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V) + \dots \right) \\
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&= (1 - \lambda) \left( \delta_t^V \left( \frac{1}{1 - \lambda} \right) + \gamma \delta_{t+1}^V \left( \frac{\lambda}{1 - \lambda} \right) + \gamma^2 \delta_{t+2}^V \left( \frac{\lambda^2}{1 - \lambda} \right) + \dots \right) \\
&= \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V
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$$g^\gamma \approx \mathbb{E} \left[ \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \hat{A}_t^{\text{GAE}(\gamma, \lambda)} \right] = \mathbb{E} \left[ \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V \right], \tag{19}$$

where equality holds when  $\lambda = 1$ .

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Reward shaping (Ng et al., 1999) refers to the following transformation of the reward function of an MDP: let  $\Phi : \mathcal{S} \rightarrow \mathbb{R}$  be an arbitrary scalar-valued function on state space, and define the transformed reward function  $\tilde{r}$  by

$$\tilde{r}(s, a, s') = r(s, a, s') + \gamma\Phi(s') - \Phi(s), \quad (20)$$

This transformation leaves the discounted advantage function  $A^{\pi, \gamma}$  unchanged for any policy  $\pi$ .

$$\sum_{l=0}^{\infty} \gamma^l \tilde{r}(s_{t+l}, a_t, s_{t+l+1}) = \sum_{l=0}^{\infty} \gamma^l r(s_{t+l}, a_{t+l}, s_{t+l+1}) - \Phi(s_t). \quad (21)$$

Let  $\Phi = V$ :

$$\sum_{l=0}^{\infty} (\gamma\lambda)^l \tilde{r}(s_{t+l}, a_t, s_{t+l+1}) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V = \hat{A}_t^{\text{GAE}(\gamma, \lambda)}. \quad (25)$$

By considering the  $\gamma\lambda$ -discounted sum of shaped rewards, we exactly obtain the generalized advantage estimators.